Dynamic Hyperspectral Pansharpening CNNs

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***Abstract*— Hyperspectral (HS) pansharpening seeks to integrate low spatial resolution HS (LRHS) images with connected panchromatic (PAN) images to produce high spatial resolution HS (HRHS) images. Traditional pansharpening convolutional neural networks (CNNs) directly map LRHS and PAN images into HRHS images under fixed network parameters, which imply static pansharpening rules. However, real-world HS data are often characterized by spatial variations, and intuitively, the pansharpening rules should be dynamic. To deal with the dilemma, in this article, we develop dynamic HS pansharpening CNNs. We first specify the concepts of dynamic pansharpening and static pansharpening. Then, we propose a learn-to-learn-oriented pansharpening CNN paradigm, which aims to learn a how-to-learn rule to produce spatially adaptive pansharpening rules and comprises three stages of preliminary fusion, scene-sensitive modulation, and spectral reconstruction. Finally, following the paradigm, we design two groups of dynamic pansharpening CNNs (DyPNNs), i.e., internal-connection-based and external-connection-based. They involve various spatial modulations, including spatial affine transform (AT), spatial dynamic convolution (DC), or improved spatial attention (SA), and, thus, consist of six specific DyPNNs: IC-AT-DyPNN, IC-DC-DyPNN, IC-SA-DyPNN, EC-AT-DyPNN, EC-DC-DyPNN, and EC-SA-DyPNN. Experimental results on several HS datasets verify the effectiveness of the proposed DyPNNs in terms of both the spatial reconstruction and spectral fidelity.**

***Index Terms*— Convolutional neural networks (CNNs), dynamic pansharpening, hyperspectral (HS) images.**

NOMENCLATURE

$\mathbf{P}$	Observed PAN image.
$\widetilde{\mathbf{H}}$	Observed LRHS image.
$\widetilde{\mathbf{H}}^{\uparrow}$	Preinterpolated version of the observed LRHS image.
$\mathbf{P}_t$	Synthetic reduced-resolution PAN image for training phase.
$\widetilde{\mathbf{H}}_t$	Synthetic reduced-resolution LRHS image for training phase.
$\widetilde{\mathbf{H}}_t^{\uparrow}$	Preinterpolated version of the synthetic reduced-resolution LRHS image for training phase.
$\widehat{\mathbf{H}}$	Pansharpened HRHS image.

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I. INTRODUCTION

IN THE design of optical remote sensing systems, one of the major considerations is the balance among the spatial resolution, the spectral resolution, and the signal-to-noise ratio (SNR) [1]. Usually, spectroscopy imaging systems are capable of densely sampling many narrow spectral bands from visible to near-infrared lights, yielding hyperspectral (HS) data with abundant spectral information, which, however, tends to degrade spatial resolutions [1], [2]. In contrast, panchromatic (PAN) imaging systems can provide single-band images with high spatial resolution. To accommodate requirements of full-resolution data in many practical remote sensing applications, such as classification [2] and target detection [3], one possible way is to reconstruct high spatial resolution HS (HRHS) images with both high spatial and high spectral resolutions from low spatial resolution HS (LRHS) images and the connected PAN images. The process is normally referred to as HS pansharpening [1], [4], [5], [6].

Over the last decade, various algorithms have been designed for HS pansharpening [1], [6], [7], [8], [9], [10], some of which could roughly be attributed to categories of component substitution (CS), multiresolution analysis (MRA), variational optimization (VO), and convolutional neural network (CNN). CS and MRA are mainly extended for classical multispectral pansharpening methods. The former usually transforms data into a specific domain where spatial component replacement is performed, typical instances of which include intensity–hue–saturation (IHS), principal component analysis (PCA), and Gram–Schmidt (GS) [6], [10], [11], [12], [13]. The latter extracts the spatial details of the PAN data and injects them into the LRHS data through the multiresolution decomposition, of which smoothing filter-based intensity modulation (SFIM), wavelet transform, and modulation transfer function generalized Laplacian pyramid (MTF-GLP) are examples [14], [15], [16], [17]. VO-based methods try to reconstruct the pansharpened image using variational theory, which usually involve ill-posed inverses to formulate pansharpening. Representative works can be found in [18], [19], and [20].

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Besides, the Bayesian estimation [21] and the matrix factorization [22], [23] can also be regarded as VO-based instances.

In fact, HS pansharpening has to deal with some special challenging problems relative to classic pansharpening [1], [4], [5], [6], for example, much wider spectrum covering of HS images compared with PAN images, and reconstructing many contiguous spectral bands in high spectral resolutions. CNNs have proven to be useful to handle these problems [4], [6], [8], [9], [10], [24], [25]. Zheng et al. [8] focused on a residual CNN structure between the pansharpened image and the LRHS image, reducing the difficulty of learning. Xie et al. [9] combined the additional priors learned by CNN and the VO-based method. He et al. [6], [10] summarized the general CNN pansharpening framework and proposed the spectrally predictive structure and the spectral-fidelity network.

As learning-based approaches, the practical implementation of HS pansharpening CNNs usually comprises two sequential phases: the reduced-resolution training and the full-resolution pansharpening [6], [10], [26], [27], although full-resolution training [28], [29] might be feasible as an additional consideration. In the reduced-resolution training phase, both the observed HS and the PAN images go through a degradation to generate resolution-reduced data, and then, a CNN model is learned with these data. In the pansharpening phase, with the stipulation of the scale invariance of the learned model, oncoming HS and PAN images are integrated under that trained CNN model to infer the desired full-resolution HS data. Although CNN-based methods have achieved significant improvements in HS pansharpening, intuitively, they ignore the consideration of spatial variations in the observed scenes that could offer potential for better performance. Spatial variation means that spectral characteristics of HS pixels vary spatially over natural scenes, and, reasonably, pansharpening should be spatially dynamic to follow the variation. More specifically, spatial variation could be attributed to two major aspects: intrascene and interscene. The intrascene variation refers to the spatial variation within an observed scene, whereas the interscene variation means the spatial variation across different scenes. Most of the traditional pansharpening CNNs did not explicitly take into account those spatial variations. First, CNN characteristics, such as weight sharing, i.e., kernel parameters are shared among various spatial positions, are usually used. This implies that CNN kernels will be forced to meet the global characteristic of an observed scene and intuitively may be inconsistent with the consideration of the position-specific intrascene variation. Undoubtedly, if those weight-sharing CNNs are deep enough with many kernels and weights and are trained with large and representative training data, they truly have the potential to adapt to spatial variation. However, the training data available in practical applications of HS pansharpening are usually rather limited. Under this situation, we may have to resort to spatially dynamic networks with structure designs specialized to spatial variation, so as to reduce the requirement of training data. Second, the data involved in the training phase and in the pansharpening phase are normally different; i.e., the training scene and the pansharpening scene are different. When we use a CNN model acquired in the training phase to enhance the data in the pansharpening scene, there may exist discrepancy brought by the interscene spatial variation.

To address the spatial variation problems mentioned above, we developed dynamic pansharpening CNNs (DyPNNs) in this article, which are able to perform spatially adaptive adjustment. More specifically, our innovative work and contributions can be identified as follows.

1) *Concepts of Dynamic Pansharpening and Static Pansharpening:* We explicitly specify the concepts of dynamic pansharpening and static pansharpening. Static pansharpening hold a pansharpening rule fixed during the pansharpening process, whereas dynamic pansharpening is able to self-adjust the pansharpening rule in terms of spatial variation.
2) *Paradigm of Learn-to-Learn-Oriented DyPNN:* This proposed paradigm is able to guide the design of specific DyPNNs. In the paradigm, a how-to-learn rule is learned from the training data and then used to automatically adjust the pansharpening rule in terms of the spatial variation of the new incoming data. We give the basic CNN formulations for the paradigm, and then, state that a DyPNN could be composed of three successive stages: preliminary fusion, scene-sensitive modulation, and spectral reconstruction.
3) *Internal Connection-Based DyPNNs (IC-DyPNNs):* Following the above paradigm, we design this group of DyPNNs. In their scene-sensitive modulation stages, several scene-sensitive modules are serialized to adjust the features adaptively through internal connections; i.e., in each scene-sensitive module, the input of a local dynamic branch (i.e., a learn-to-learn subnetwork) is exclusively from inside the main trunk. The group contains three specific CNNs, including affine transform IC-DyPNN (IC-AT-DyPNN), dynamic convolutional IC-DyPNN (IC-DC-DyPNN), and spatial attention IC-DyPNN (IC-SA-DyPNN).
4) *External Connection-Based DyPNNs (EC-DyPNNs):* These groups of CNNs are also designed by following the proposed learn-to-learn paradigm. In each scene-sensitive module of EC-DyPNNs, the main trunk is also accompanied by a local dynamic branch (i.e., a learn-to-learn subnetwork), but the dynamic branch inputs data from outside the main trunk. This group also contains three specific CNNs, including EC-AT-DyPNN, dynamic convolutional EC-DyPNN (EC-DC-DyPNN), and EC-SA-DyPNN.

The rest of this article is organized as follows. Section II specifies the concepts of the dynamic pansharpening and the static pansharpening, and then a paradigm for learn-to-learn DyPNNs. Section III presents the details of the developed IC-DyPNNs and EC-DyPNNs. The experimental results and analyses are presented in Section IV. Finally, the conclusion is drawn in Section V.

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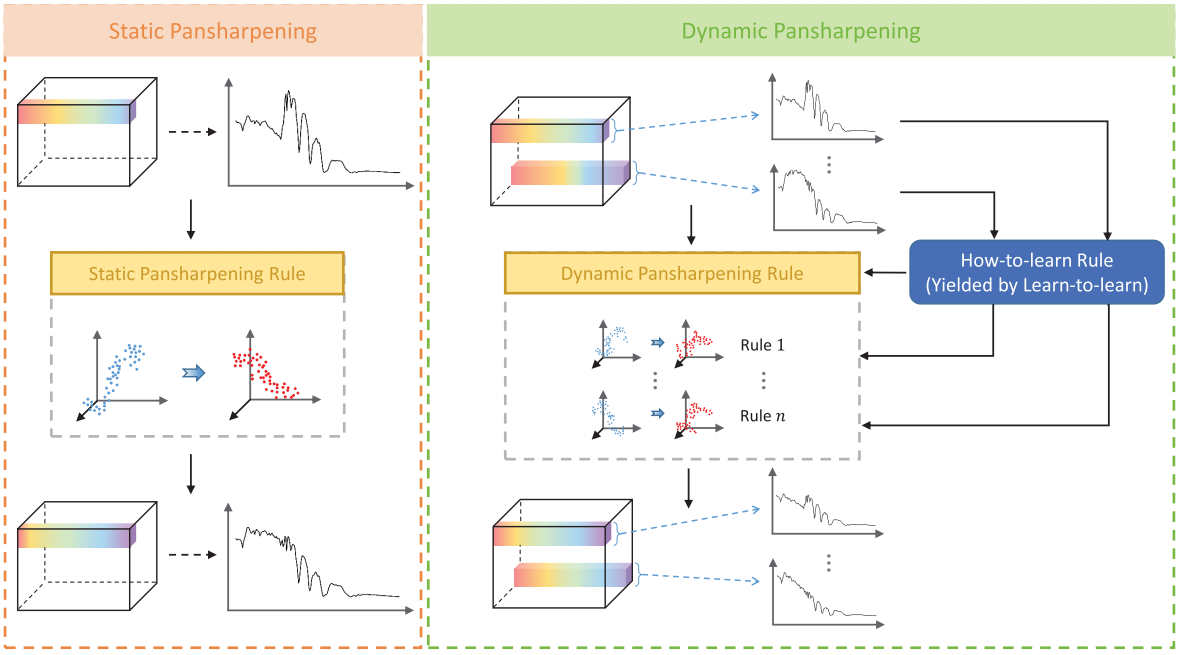

Fig. 1. Basic threads of static pansharpening and dynamic pansharpening.

II. Dynamic Pansharpening

As discussed in Section I, the motive to our work stems from an intuitive observation that spectral characteristics of HS pixels on nature scenes vary spatially. Such spatial variations, including intrascene variation and interscene variation, are what most of the existing CNN-based pansharpening methods fail to explicitly take into consideration.

Our work seeks to develop innovative DyPNNs to cope with the spatial variation problem. In this section, we first specify the concepts of the dynamic pansharpening and the static pansharpening, especially a learn-to-learn-oriented dynamic pansharpening paradigm. Then, the basic formulation for the CNN-based dynamic pansharpening is described in detail. For clear description, major notations used hereafter are listed in the Nomenclature.

A. Static Pansharpening and Dynamic Pansharpening

Here, we specify the concepts of static pansharpening and dynamic pansharpening, which have not been explicitly mentioned in the previous literature.

1) *Static Pansharpening:* Pansharpening via a static rule that is not changed or adjusted during the pansharpening process. As shown in Fig. 1 (left), under the static pansharpening rule, the mapping from input data to pansharpened data is globally fixed in spite of any spatial variations.
2) *Dynamic Pansharpening:* Pansharpening by means of a dynamic pansharpening rule that can follow the spatial variations of the processed data. In contrast to static pansharpening, dynamic pansharpening offers pansharpening mappings that vary depending on spatial variations, as shown in Fig. 1 (right).

In our work, we use a learn-to-learn strategy [30], [31] to build the dynamic mappings desired in DyPNNs. *Learn-to-learn in our CNN design means learning how to learn pansharpening. More specifically, it aims at learning how to generate dynamic pansharpening models/rules, i.e., learning a how-to-learn rule to yield pansharpening rules spatial adaptive to new data.* As shown in Fig. 1 (right), in our dynamic pansharpening, a how-to-learn rule is first trained by a learn-to-learn process, which establishes a mapping from input data to pansharpening rules. Once the how-to-learn rule is determined, it can be applied on the oncoming data, and the data would drive the how-to-learn rule to produce dynamic pansharpening rules, which are spatially adaptive to the data. By contrast, normal static pansharpening CNNs first train a pansharpening model, and then, this model is directly used as the pansharpening rule in the pansharpening phase to infer the final pansharpening result.

In the following, we pose CNN formulations inspired by the above dynamic pansharpening thread.

B. CNN Formulations of Dynamic Pansharpening

Normally, pansharpening CNN methodology is learning-based, and its practical implementation is divided into two sequential phases: the training phase and the pansharpening phase, which can be formulated as follows:

$$\boldsymbol{\theta}^{\text{opt}} = \arg\min_{\theta} \mathcal{L}(\Phi(\widetilde{\mathbf{H}}_t^{\uparrow}, \mathbf{P}_t; \boldsymbol{\theta}), \widetilde{\mathbf{H}}) \tag{1}$$

and

$$\widehat{\mathbf{H}} = \Phi(\widetilde{\mathbf{H}}^{\uparrow}, \mathbf{P}; \boldsymbol{\theta}^{\text{opt}}) \tag{2}$$

respectively, where $\Phi(\cdot)$ represents the CNN panshrapening mapping from input data (preinterpolated LRHS images + PAN images) to full-resolution images with CNN parameters

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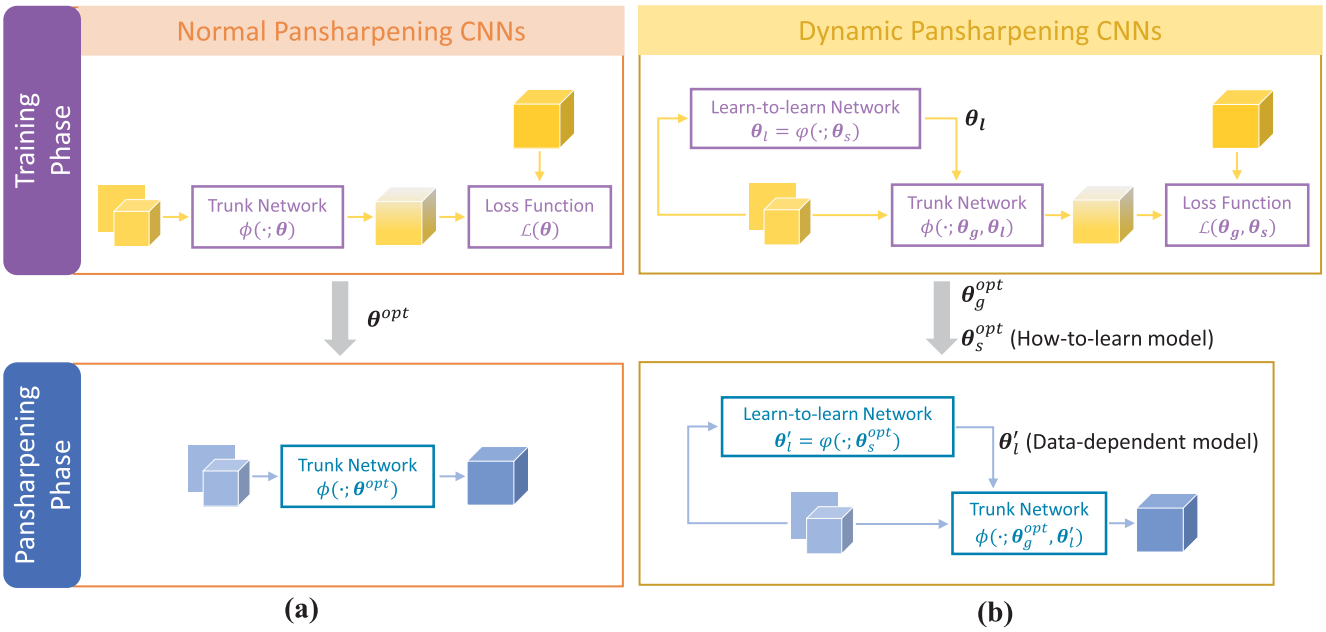

Fig. 2. Schematics of learning-based pansharpening CNNs. (a) Normal pansharpening CNNs. (b) DyPNNs.

$\boldsymbol{\theta}$, and $\mathcal{L}$ denotes the pansharpening loss in the training phase, which imposes constraints on the differences between pansharpened full-resolution images and referenced ground truth. The formulations of (1) and (2) for normal pansharpening CNNs can be graphically shown in Fig. 2(a). As shown, a pansharpening model/rule $\boldsymbol{\theta}^{\text{opt}}$ is learned in the training phase. During the succeeding pansharpening phase, the learned pansharpening rule $\boldsymbol{\theta}^{\text{opt}}$ is applied on the incoming data to infer final HRHS images. That is, the pansharpening CNN model is static during the pansharpening process.

In contrast to the normal HS pansharpening, dynamic HS pansharpening works in a learn-to-learn manner to enable dealing with spatial variations, and, thus, could have mathematical representations as follows:

$$\boldsymbol{\theta}_g^{\text{opt}}, \boldsymbol{\theta}_s^{\text{opt}} = \arg\min_{\theta_g,\theta_s} \mathcal{L}(\Phi(\widetilde{\mathbf{H}}_t^{\uparrow}, \mathbf{P}_t; \boldsymbol{\theta}_g, \boldsymbol{\theta}_l = \varphi(\mathbf{S}_t; \boldsymbol{\theta}_s)), \widetilde{\mathbf{H}}) \quad (3)$$

and

$$\widehat{\mathbf{H}} = \Phi(\widetilde{\mathbf{H}}^{\uparrow}, \mathbf{P}; \boldsymbol{\theta}_g^{\text{opt}}, \boldsymbol{\theta}_l = \varphi(\mathbf{S}; \boldsymbol{\theta}_s^{\text{opt}})) \quad (4)$$

where $\boldsymbol{\theta}_g$ denotes the CNN parameters of the trunk subnetwork of the pansharpening CNN, $\boldsymbol{\theta}_l$ represents the dynamic parameters spatially adaptive, $\boldsymbol{\theta}_s$ stands for the CNN parameters of the learn-to-learn subnetwork, which determines $\boldsymbol{\theta}_l$, and $\mathbf{S}_t$ and $\mathbf{S}$ refer to the data inside the pansharpening CNN to feed into the learn-to-learn subnetwork in the training phase and in the pansharpening phase, respectively. Equations (3) and (4) formulating DyPNNs can be graphically shown in Fig. 2(b). As shown, once parameters of the learn-to-learn subnetwork $\boldsymbol{\theta}_s$ are learned in the training phase, the how-to-learn rule $\boldsymbol{\theta}_s^{\text{opt}}$ is determined. Then, in the succeeding pansharpening phase, spatially varying data would drive the learn-to-learn subnetwork to yield spatially varying pansharpening rule $\boldsymbol{\theta}_l$, which collaborates with the learned main trunk parameters $\boldsymbol{\theta}_g^{\text{opt}}$ to perform on oncoming data so as to accomplish the final pansharpening.

III. Two Groups of Dynamic Pansharpening CNNs

In this section, specifically, we develop two groups of pansharpening CNN instantiations following our paradigm proposed in Section II-B. The first group comprises three IC-DyPNNs: spatial IC-AT-DyPNN, spatial IC-DC-DyPNN, and improved IC-SA-DyPNN, while the second group is composed of three EC-DyPNNs: EC-AT-DyPNN, EC-DC-DyPNN, and EC-SA-DyPNN. Fig. 3 graphically illustrates the three major stages composing our DyPNNs: preliminary fusion, which is responsible for rough data preintegrating, scene-sensitive modulation, which enables the intermediate data to adaptively follow spatial variation, and spectral reconstruction, which enforces the final data with the spectral fidelities.

A. IC-DyPNNs

The overall structure for the proposed IC-DyPNNs is shown in Fig. 4. In the preliminary fusion stage, an IC-DyPNN takes the PAN and the upsampled LRHS images as its inputs and feeds them into the double gated recurrent module (DGRM), yielding the preliminary fused data. Subsequently, in the scene-sensitive modulation stage, several scene-sensitive module block are serialized to adjust the features adaptively through the internal connections. In each scene-sensitive module, the input of a local dynamic branch (i.e., a learn-to-learn subnetwork) is exclusively from inside the main trunk, and therefore, we call this group of DyPNNs internal connection-based networks. Finally, in the spectral reconstruction stage, a spectral reconstruction subnetwork is carried out to restore the spatial details for each band and generate the pansharpening result.

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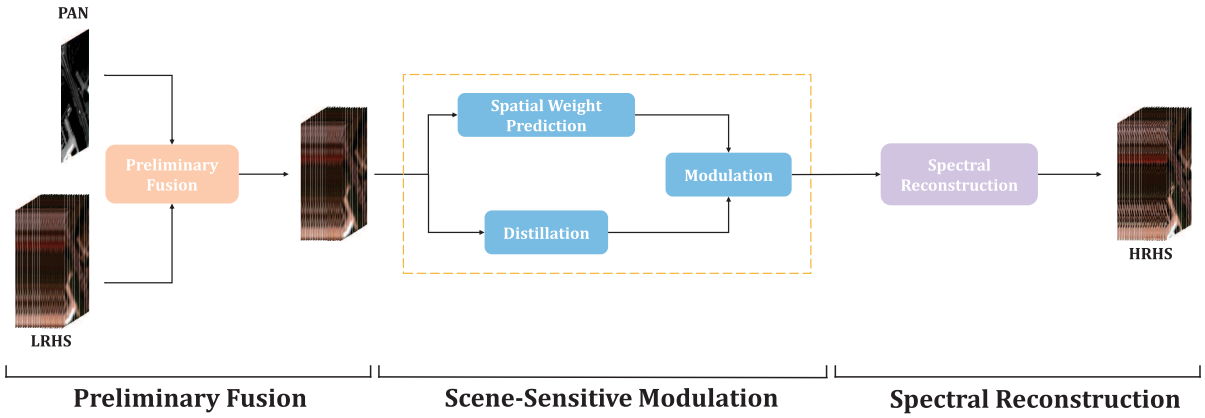

Fig. 3. Proposed three-stage HS pansharpening framework.

1) Preliminary Fusion: Most of the existing pansharpening CNNs input the PAN image as an additional band of the upsampled LRHS image and then treating each band equally. However, since the characteristics of the PAN image and the LRHS image are significantly different, such a way may be unable to fully extract the information in these two images. One solution is to process the PAN image and the LRHS image separately [10], which, however, neglects the inherent connection between these two images. To solve the problem, we construct a DGRM module here to extract the features of the PAN and LRHS images in a compromise manner, which introduces the gated recurrent unit (GRU) [32] for the information interaction through the gated mechanism.

As shown in Fig. 4, DGRM is built upon stacking several GRU units, each of which aims to update the current hidden state by fusing the current input and the previous hidden state, with two gated signals indicating their ratio. More specifically, a GRU unit combines the previous hidden state $\boldsymbol{h}_{t-1}$ at the step $t-1$ and the current input $\mathbf{x}_t$ at the step t to yield the grated signals $\boldsymbol{u}$ and $\boldsymbol{v}$ at first. Subsequently, these two grated signals will be applied to control the information flows by reweighting to update the current hidden state $\boldsymbol{h}_t$. The mathematical model can be formulated as follows:

$$\begin{aligned}\boldsymbol{u} &= \sigma(\mathbf{W}_u[\boldsymbol{h}_{t-1}, \mathbf{x}_t] + b_u)\\ \boldsymbol{v} &= \sigma(\mathbf{W}_v[\boldsymbol{h}_{t-1}, \mathbf{x}_t] + b_v)\\ \hat{\boldsymbol{h}}_t &= \text{relu}(\mathbf{W}_h[\boldsymbol{v} \odot \boldsymbol{h}_{t-1}, \mathbf{x}_t] + b_h)\\ \boldsymbol{h}_t &= \boldsymbol{u} \odot \hat{\boldsymbol{h}}_t + (1-\boldsymbol{u}) \odot \boldsymbol{h}_{t-1}\end{aligned} \tag{5}$$

where $\mathbf{W}_u$, $\mathbf{W}_v$, and $\mathbf{W}_h$ are the weights of different convolution layers, b_u, b_v, and b_h are bias terms, $[\cdot,\cdot]$ denotes the concatenation operation, and relu$(\cdot)$ denotes the rectified linear unit. DGRM consists of two branches. One branch takes the PAN image as its input; the other branch takes the LRHS image as its input. The outputs of the first layer of GRUs in the PAN and LRHS branches are designed to cross connect to the second layer of GRUs to function as their previous hidden states to perform interaction between the two branches, as shown in the DGRM structure in Fig. 4. In this way, the PAN branch and the LRHS branch guide each other during the DGRM feature extraction: the spatial details in the PAN image can guide the detail feature extraction of the LRHS branch, while the global object contours in the LRHS image can be helpful for locating global features in the PAN image.

2) Scene-Sensitive Modulation: In our IC-DyPNNs, this stage performs the core function of spatially adaptive adjustment in an internal connection fashion. As shown in Fig. 4, it is composed of a sequence of scene-sensitive modules. Each scene-sensitive module detects the spatial variations of its input and accordingly tunes its output along the spectral dimension. The cascaded tunings of multiple scene-sensitive modules enable the effective spatial modulation.

Each scene-sensitive module comprises a main trunk and a dynamic branch. The dynamic branch uses a multiscale dilation convolution block (MSDB) to perform as a learn-to-learn subnetwork. That is, once the weight parameters of MSDB are determined with a training process, a rule to generate dynamic parameters, which are spatially adaptive to its input data, is obtained. In the main trunk of a scene-sensitive module, a convolutional block (CBlock) is employed to fulfill feature extraction with several convolution layers. Then, the feature maps extracted from the main trunk are input to a modulating operation unit to be processed with the dynamic parameters from MSDB, fulfilling a spatial modulation or adjustment. The modulating operation can be performed in three manners in our design: spatial AT, spatial dynamic convolution (DC), and improved SA, which are connected to our three specific IC-DyPNN methods: IC-AT-DyPNN, IC-DC-DyPNN, and IC-SA-DyPNN.

It is noteworthy that the dynamic branch serves as the learn-to-learn subnetwork to yield dynamic parameters sensitive to its input data. This data dependence of dynamic parameters is associated with the feature extraction capability of the branch, i.e., how to capture scene characteristics sufficiently. Since various kinds of ground objects with different spatial structures may occur in the scenes, receptive fields covered multiple scales are desired to enhance such a feature extraction capability. Dilation convolution [33], [34], [35] is introduced for the design of our MSDB. As shown in the structure in Fig. 4, MSDB transfers the input feature maps into three pathways, and each pathway consists of several consecutive dilation convolution layers. Note that the dilation convolution layers in different pathways are with different dilation

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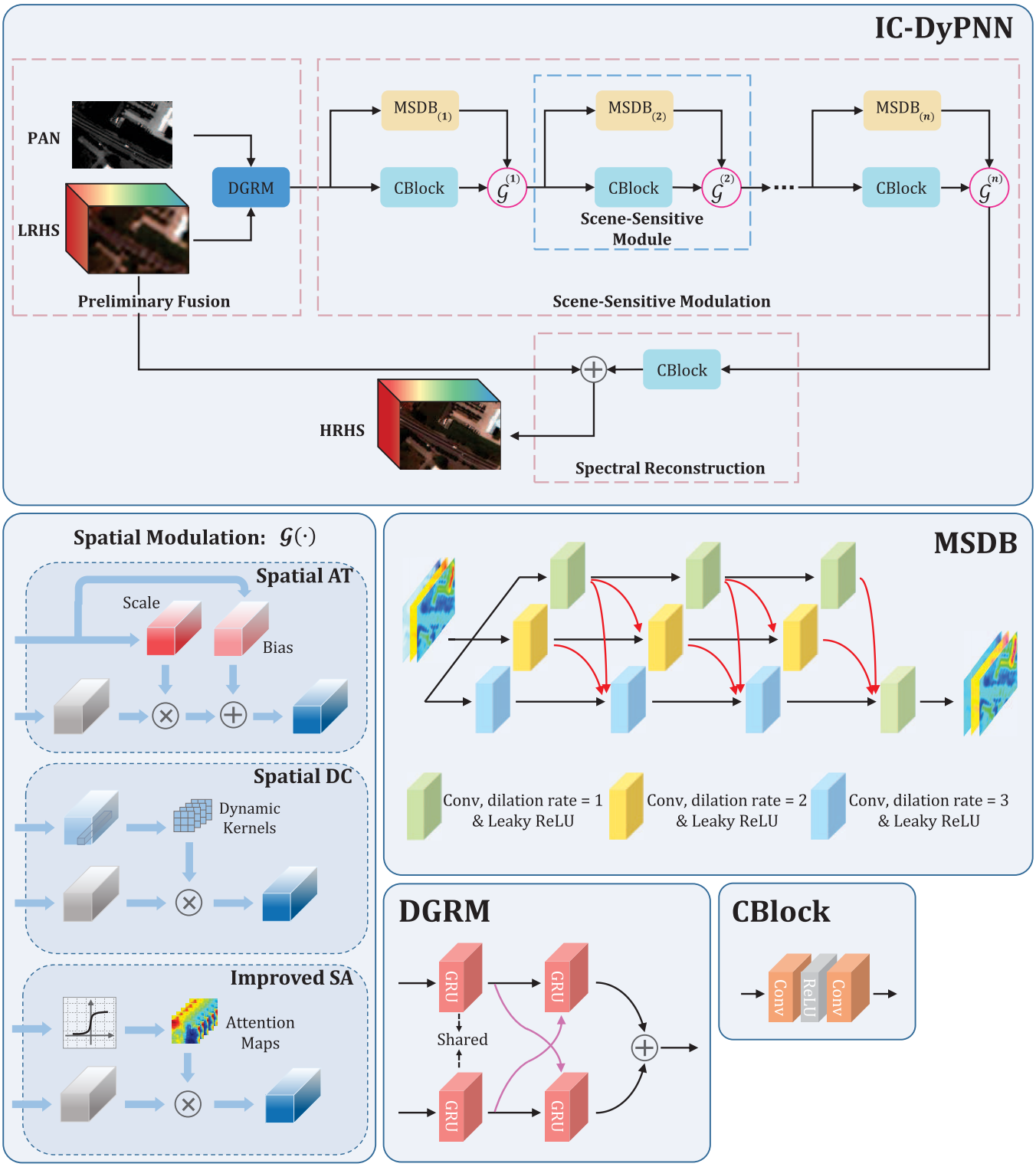

Fig. 4. Overall structure of the proposed IC-DyPNNs.

rates, and each one takes the cross feature maps (i.e., the feature maps come from all the upper pathways) as its input. Consequently, multiscale receptive fields are implemented by crossing different pathways. Empirically, the dilation rates for these three pathways could be set as 1, 2, and 3, respectively, and convolution kernels could be with the size of 3×3; thus, MSDB can cover the receptive field range of 7×7–19×19.

As aforementioned, after obtaining dynamic parameters with the dynamic branch, the spatial modulation of the feature maps in the main trunk can be accomplished by a suitable modulating operation unit $\mathcal{G}(\cdot)$. $\mathcal{G}(\cdot)$ can be set to $\mathcal{G}_{\mathrm{AT}}(\cdot)$ as in (6), $\mathcal{G}_{\mathrm{DC}}(\cdot)$ as in (7), or $\mathcal{G}_{\mathrm{SA}}(\cdot)$ as in (8).

a) Spatial affine transform: A mechanism that can be used to design the modulating operation unit $\mathcal{G}(\cdot)$ is spatial

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AT [36], which is originally proposed to transform the semantic probability map to the target feature map through an affine transformation, where the semantic categorical prior is utilized to restore texture discriminatively. In our design, the mathematical representation of the spatial AT process is of the form

$$\begin{aligned} \mathcal{G}_{\mathrm{AT}}(\mathbf{F}) &= \boldsymbol{\gamma} \odot \mathbf{F} + \boldsymbol{\beta} \\ (\boldsymbol{\gamma}, \boldsymbol{\beta}) &= \mathcal{M}_{\mathrm{AT}}(\mathbf{C}) \end{aligned} \tag{6}$$

where $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ denote the scale and bias parameters of the AT, $\mathbf{F}$ represents the feature maps from the main trunk, $\mathbf{C}$ represents the input of the dynamic branch MSDB, and $\mathcal{M}_{\mathrm{AT}}$ stands for the mapping from the input of MSDB to modulation parameters. Therefore, modulation parameters $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ refine the semantic information from the input of the dynamic branch with the mapping $\mathcal{M}_{\mathrm{AT}}$. The affine transformation in (6) is spatialwise. By this way, the spatial AT provides a way to modulate each pixel according to its semantic priors by scaling and shifting feature maps.

b) Spatial dynamic convolution: DC [37], which we also can utilize for a learn-to-learn design, adaptively applies convolution kernels condition on the related input, which is different from the standard convolution, where the weights of the kernel are fixed. In the design of our spatial DC, the kernels are position-specific on intermediate feature maps. For each position (i, j) of a feature map $\mathbf{F}$, the DC can be formulated as follows:

$$\begin{aligned} \mathcal{G}_{\mathrm{DC}}(\mathbf{F}_{i,j}) &= \boldsymbol{\delta}_{i,j} * \mathbf{F}_{i,j} \\ \boldsymbol{\delta}_{i,j} &= \mathcal{M}_{\mathrm{DC}}(\mathbf{C}) \end{aligned} \tag{7}$$

where $*$ denotes the convolution operation, and $\boldsymbol{\delta}_{i,j}$ is the DC kernel at position (i, j) generated with the mapping $\mathcal{M}_{\mathrm{DC}}(\cdot)$, where the dynamic branch serves as a filter-generating network. Unlike the spatial AT, which only works on each pixel, the spatial DC takes the pixel neighbors into consideration.

c) Improved spatial attention: SA explores how to selectively focus on spatially important or salient parts of the data [38]. In the design of our improved SA, dynamic branches MSDBs (i.e., learn-to-learn subnetworks) serve as the SA modules to produce SA maps, and then, the feature maps are used to tune the intermediate feature maps output from the main trunk, which is different from the original SA where the attention maps yielded by the attention modules are used to directly tune the inputs of the attention modules. Our improved SA can be formulated as follows:

$$\begin{aligned} \mathcal{G}_{\mathrm{SA}}(\mathbf{F}) &= \boldsymbol{\alpha} \odot \mathbf{F} \\ \boldsymbol{\alpha} &= \sigma(\mathcal{M}_{\mathrm{SA}}(\mathbf{C})) \end{aligned} \tag{8}$$

where $\boldsymbol{\alpha}$ represents the SA map generated by utilizing the interspatial relationship of the input, $\odot$ denotes the elementwise multiplication, $\mathcal{M}_{\mathrm{SA}}(\cdot)$ is the mapping function implemented by a specific CNN, and $\sigma(\cdot)$ is the sigmoid function. The values of the SA map $\boldsymbol{\alpha}$ are between 0 and 1, which can be consider as a probability map that indicates the importance of each spatial position. When multiplying by $\boldsymbol{\alpha}$, the important positions in $\mathbf{F}$ will be highlighted, and those unimportant will be suppressed. From the point of view of the learn-to-learn strategy, the SA assigns an input-based weight factor for each pixel to adjust its impact on the output according to its own characteristics.

3) Spectral Reconstruction: After the scene-sensitive modulation stage, a spectral reconstruction stage is involved to spectrally rescale the feature maps and strengthen the spectral fidelity of the final pansharpening result. The spectral reconstruction stage contains two consecutive convolution layers and generates the final pansharpened image within the detail injection framework

$$\widehat{\mathbf{H}} = \widetilde{\mathbf{H}}^{\uparrow} + \mathcal{J}(\widetilde{\mathbf{H}}^{\uparrow}, \mathbf{P}) \tag{9}$$

where $\mathcal{J}(\cdot)$ denotes the detail mapping functions of IC-DyPNNs. In this way, IC-DyPNNs only needs to reconstruct the spatial details of the targeted pansharpened image rather than the whole image, relieving the difficulty of image restoration.

B. EC-DyPNNs

Besides the group of IC-DyPNNs, in this section, we develop a group of EC-DyPNNs. In contrast to IC-DyPNNs, a dynamic branch (i.e., a learn-to-learn subnetwork) inputs data from outside the main trunk in each scene-sensitive module of EC-DyPNNs.

Fig. 5 shows the overall architecture for the proposed EC-DyPNNs. As seen, the architecture is similar to that for IC-DyPNNs, which also consists of three sequential stages. In the preliminary fusion stage, input PAN and LRHS images are merged by means of the proposed DGRM, obtaining the preliminary fused data. The scene-sensitive modulation stage starts with a scene-sensitive module using in IC-DyPNNs, but followed by consecutive redesigned scene-sensitive modules. Different from those modules in IC-DyPNNs, which are independent of each other, the redesigned scene-sensitive modules in EC-DyPNNs take the previous multiple scene features with various scales as the input of the dynamic branch through short-cut connections. Here, two convolution layers are applied to aggregate these features before feeding them into MSDB. In this way, multiscale scene information can be gradually accumulated in the dynamic branch, and thus, more dynamic parameters for the succeeding spatial modulation can be obtained. Three spatial modulation patterns, including $\mathcal{G}_{\mathrm{AT}}(\cdot)$, $\mathcal{G}_{\mathrm{DC}}(\cdot)$, and $\mathcal{G}_{\mathrm{SA}}(\cdot)$ used in IC-DyPNNs, can also be utilized to tune the main trunk with dynamic parameters from the dynamic branch, thus forming three specific EC-DyPNNs: EC-AT-DyPNN, EC-DC-DyPNN, and EC-SA-DyPNN. After yielding spatial modulating features in the scene-sensitive modulation stage, a spectral reconstruction stage is carried out for the aim of rescaling spectral bands and enhancing the spectral restoration of the final pansharpening result.

C. Summary

Following our learn-to-learn dynamic pansharpening paradigm, we construct several specific DyPNNs through internal connection or external connection, while using three specific modulation operations: spatial AT, spatial DC, or improved SA, to address spectral variation. In each of

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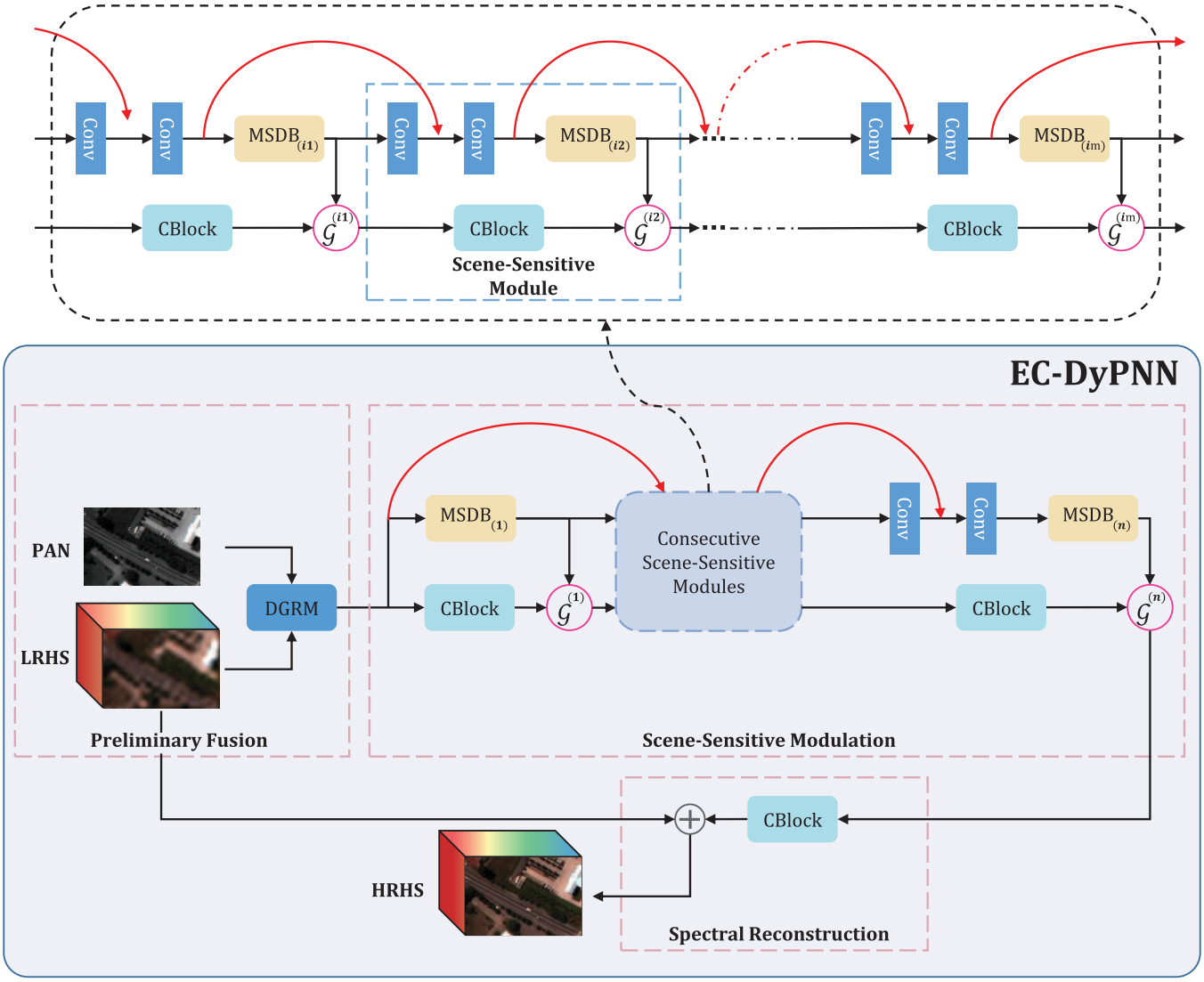

Fig. 5. Overall structure of the proposed EC-DyPNNs.

DyPNNs, the dynamic branch of a scene-sensitive module functions as a learn-to-learn subnetwork, mapping its input data to dynamic parameters, which represents dynamic pansharpening rule. Once the parameters of the dynamic branches are learned in the training stage (i.e., how-to-learn rule is determined), it can be applied on the oncoming data in the pansharpening stage, and the data would drive the how-to-learn rule to produce position-specific dynamic modulation parameters. Then, the dynamic parameters are used to spatially modulate the connected main trunk. In this way, the intrascene variation problem is alleviated. The interscene variation problem could be considered based on the intrascene variation problem. Interscene variation is the problem of scene differences, which is, in fact, to pansharpen a new set of pixels with a trained model. As mentioned above, under the learn-to-learn mechanism, the pansharpening rule is position-specific for the incoming data in the pansharpening stage, which implies that new scene would lead to automatic spatial adjustment of the pansharpening rule.

IV. Experimental Results and Analysis

In this section, we first provide a brief description of the datasets used for performance measures. Then, evaluation metrics and compared pansharpening methods are depicted, as well as related experimental settings. Finally, numeric and visual experimental results are presented, together with the corresponding qualitative and quantitative analyses.

A. Experimental Datasets

Experiments to assess the performance of the proposed method are carried out on four datasets: Washington DC Mall, Pavia University, Houston, and Merced. The characteristics of these datasets are summarized below and can be found in Table I.

1) *Washington DC Mall Dataset:* This dataset collected by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor is composed of an urban scene in Washington DC Mall. The collected HS image consists of 210 bands in the spectral range of 0.4–2.5 μm, which has a ground sampling distance (GSD) of 2.5 m. Several bands have been removed because of the opaque atmosphere, and 191 bands are used in our experiments.
2) *Pavia University Dataset:* This dataset was acquired by the Reflective Optics System Imaging Spectrometer

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TABLE I
CHARACTERISTICS OF DATASETS USED FOR PERFORMANCE MEASURES

Datasets	Test Data Size	Spectral Range	GSD	Instrument
Washington DC Mall	HS 50 ×50×191 PAN 250×250	0.4-2.5 μm	HS 2.5 m	HYDICE
Pavia University	HS 32×32×103 PAN 160×160	0.43-0.838 μm	HS 1.3 m	ROSIS
Houston	HS 60×60×144 PAN 300×300	0.364-1.046 μm	HS 2.5 m	CASI-1500
Merced	HS 60×60×134 PAN 180×180	0.4-2.5 μm	HS 30 m PAN 10 m	Hyperion and ALI

TABLE II
QUALITY METRICS FOR THE WASHINGTON DC MALL DATASET

Ours	*IC-SA-DyPNN*	*IC-AT-DyPNN*	*IC-DC-DyPNN*	*EC-SA-DyPNN*	*EC-AT-DyPNN*	*EC-DC-DyPNN*
CC(↑)	0.9640	0.9621	**0.9649**	0.9624	0.9647	0.9644
SAM(↓)	3.6671	3.6783	3.6671	3.7110	**3.6452**	3.6581
ERGAS(↓)	3.1224	3.2062	**3.0640**	3.1764	3.0710	3.0815
RMSE(↓)	**0.0141**	0.0143	0.0142	0.0143	**0.0141**	0.0143
$Q2^n$(↑)	0.9452	0.9409	**0.9456**	0.9437	0.9453	0.9446
Existing	Interpolation	GSA	SFIM	CNMF	BayesSparse	HySure
CC(↑)	0.8209	0.9148	0.9195	0.8834	0.9377	0.9140
SAM(↓)	7.0002	6.9698	6.6154	8.3770	5.7578	6.4387
ERGAS(↓)	6.3669	4.6337	4.5755	5.4528	4.0335	4.8437
RMSE(↓)	0.0331	0.0232	0.0237	0.0271	0.0191	0.0221
$Q2^n$(↑)	0.6617	0.8648	0.8534	0.8275	0.9039	0.8647
Existing	APNN	DRPNN	HyperPNN1	HyperPNN2	HSpeNet1	HSpeNet2
CC(↑)	0.8981	0.9565	0.9482	0.9536	0.9570	0.9574
SAM(↓)	4.9893	3.9835	3.9197	3.9235	3.7863	3.8467
ERGAS(↓)	6.7592	3.4568	4.0301	3.5807	3.2985	3.2943
RMSE(↓)	0.0167	0.0144	0.0145	0.0145	0.0155	0.0154
$Q2^n$(↑)	0.8107	0.9322	0.9198	0.9331	0.9350	0.9325

(ROSIS) sensor over Pavia University in Italy, which provides an HS image with 115 bands covering the spectral range of 0.43–0.838 μm and a GSD of 1.3 m. Water absorption bands and damaged bands are discarded, so that only 103 bands are used in our experiments.

3) *Houston Dataset:* This dataset covers the university campus and its neighboring urban area in Houston. It was acquired by the Compact Airborne Spectrographic Imager (CASI)-1500 sensor with a GSD of 2.5 m and contains an HS image characterized by 144 bands in the spectral range of 0.364–1.046 μm.
4) *Merced Dataset:* Covering some vegetational areas of Merced, California, this dataset was captured by the Earth Observing One (EO-1) satellite. Specifically, with a spectral spanning from 0.4 to 2.5 μm and a GSD of 30 m, the HS image of the dataset comprises 242 bands and 134 bands of which are selected for experiments; the connected PAN image is with a GSD of 10 m.

Since there are no available high-resolution reference images, the training process should be performed on the reduced resolution. That is, pansharpening CNNs in comparison should be trained in a reduced-resolution stage, holding the assumption of invariance of quality among resolution scales when pansharpening original observed images. In this case, synthetic LRHS and PAN images are generated from original observed images according to the Wald's protocol [39], usually by Gaussian blurring and downsampling. Subsequently, PAN images and preinterpolated LRHS images at the scale of PAN images using a polynomial kernel [40] can be applied for pansharpening, while original observed HS images serve as referenced ground truths.

B. Metrics and Compared Methods

The availability of reference images makes it possible to carry out quality assessments by means of proper metrics. Here, five often-used evaluation metrics are considered in our experiments, including cross correlation (CC) [1], spectral angle mapper (SAM) [41], *Erreur Relative Globale Adimensionnelle de Synthèse* (ERGAS) [42], root-mean-squared error (RMSE) [1], and an extension of the universal image quality

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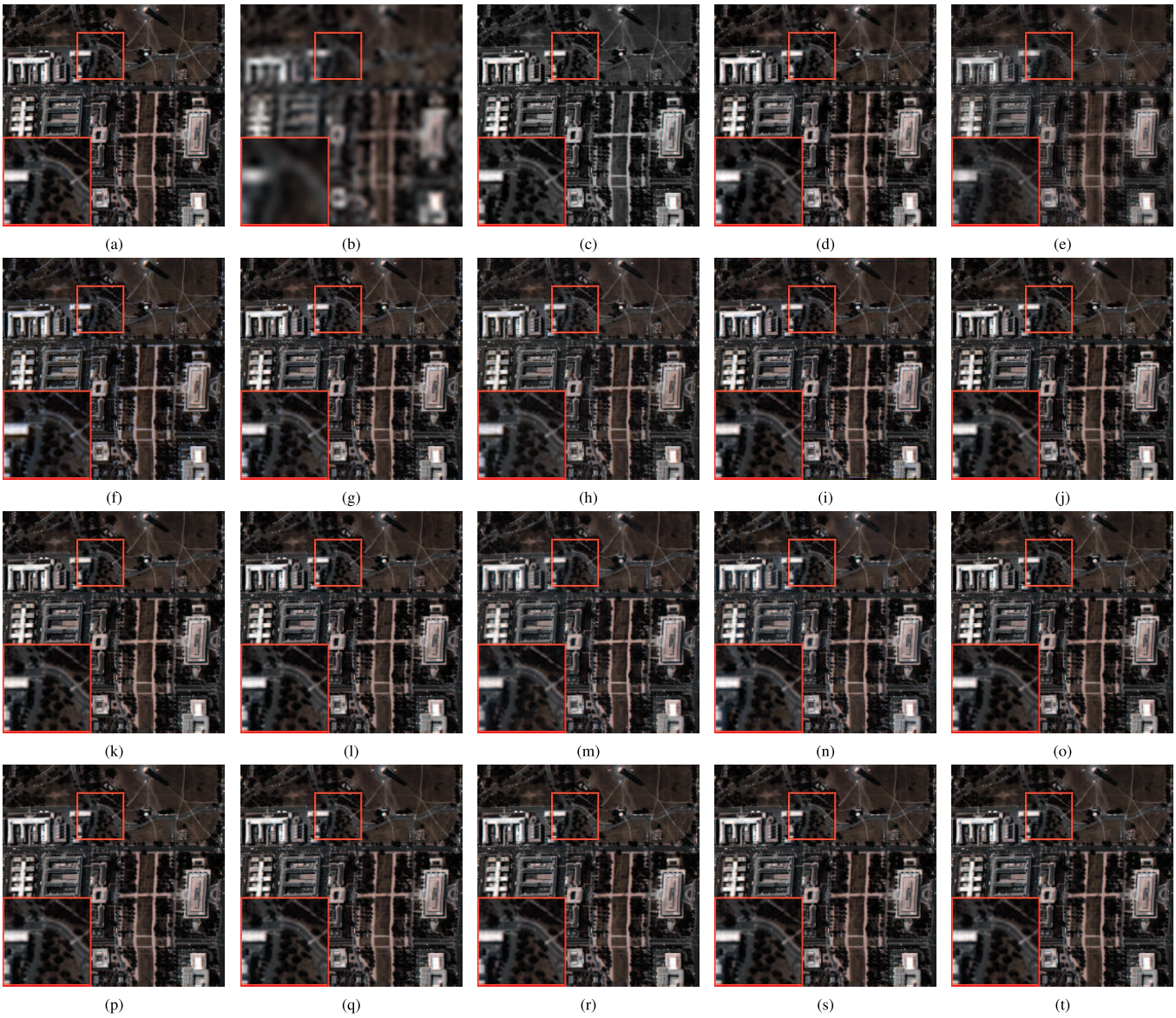

Fig. 6. Pansharpening results of various methods on the Washington DC Mall dataset. (a) Ground truth. (b) Interpolation. (c) PAN. (d) GSA. (e) SFIM. (f) CNMF. (g) BayesSparse. (h) HySure. (i) APNN. (j) DRPNN. (k) HyperPNN1. (l) HyperPNN2. (m) HSpeNet1. (n) HSpeNet2. (o) IC-SA-DyPNN. (p) IC-AT-DyPNN. (q) IC-DC-DyPNN. (r) EC-SA-DyPNN. (s) EC-AT-DyPNN. (t) EC-DC-DyPNN.

index ($Q2^n$) [43]. Among them, CC characterizes the spatial distortion, while SAM is a spectral metric, and the rest offer a global quality evaluation.

In our experiments, several state-of-the-art pansharpening methods are compared with our DyPNNs, including traditional adaptive Gram-Schmidt (GSA) [44], SFIM [14], coupled nonnegative matrix factorization (CNMF) [23], Bayesian sparse (BayeSparse) [45], and hyperspectral superresolution (HySure) [46], as well as CNN-based methods comprising target-adaptive pansharpening neural network (APNN) [47], deep residual pansharpening neural network (DRPNN) [48], spectrally predictive hyperspectral pansharpening CNNs (HyperPNNs) [6], and HS spectral-fidelity networks (HSpeNets) [10]. GSA is one of the typical CS methods characterized by high spatial fidelity, while SFIM belongs to MRA class and holds an appreciable spectral consistency. CNMF, BayeSparse, and HySure are representative VO-based methods, which achieve a good pansharpening performance. All implementations of these traditional methods follow the code in [1]. Among the CNN-based methods, APNN and DRPNN were originally proposed for multispectral pansharpening, and here, we reuse them for HS pansharpening. To ensure complete reproductions, all the CNN-based methods follow the same convolutional layer settings as in original papers. In addition, for the sake of fairness, we keep the same training settings, such as optimizer and the initial learning rate for all methods, and retrain them using the same training samples.

C. Implementation Details

All the experiments are conducted on an NVIDIA GeForce RTX 2060 SUPER GPU, with Intel Core[1] i7-10700F CPU

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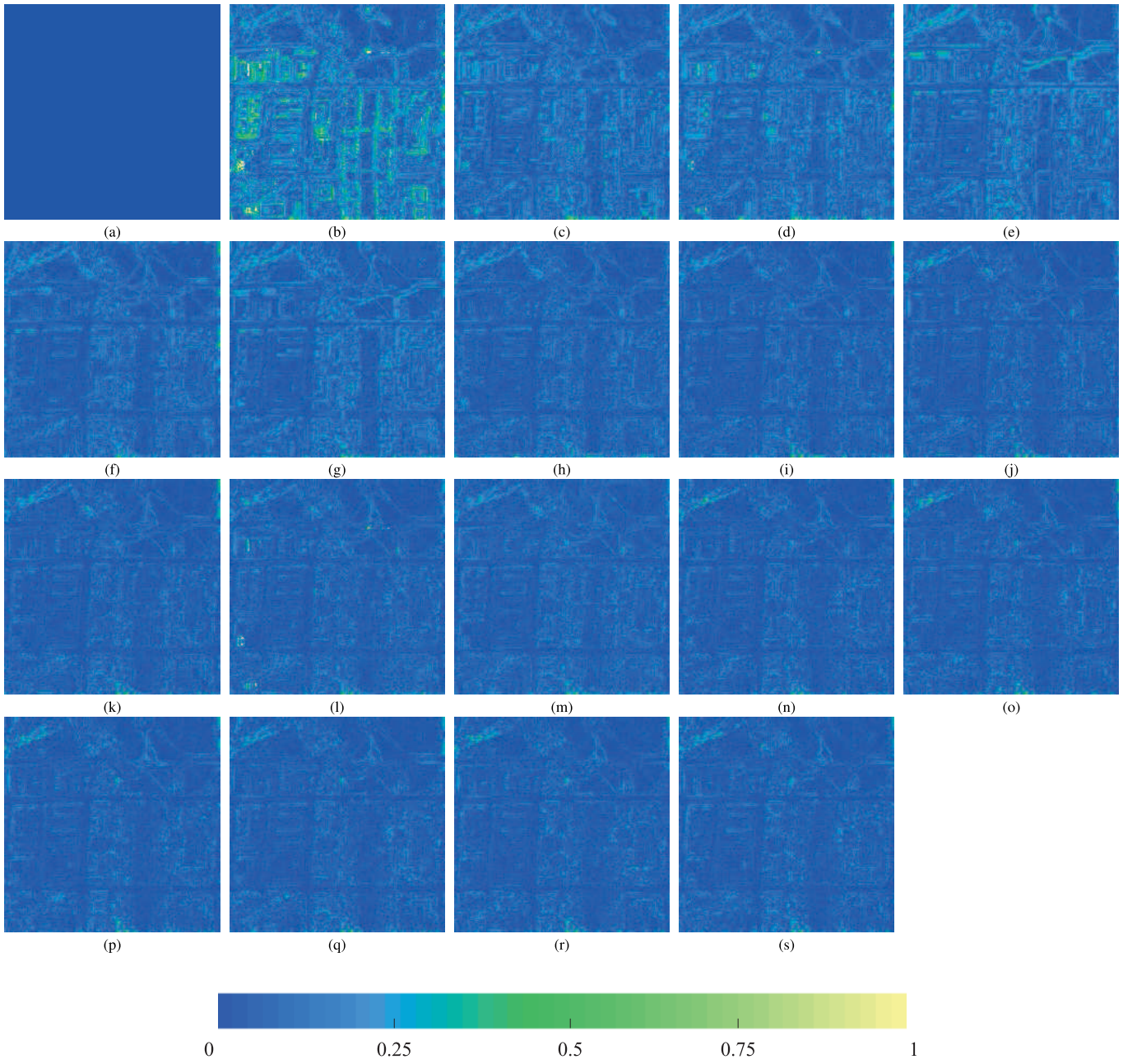

Fig. 7. Error images of various methods on the Washington DC Mall dataset. (a) Ground truth. (b) Interpolation. (c) GSA. (d) SFIM. (e) CNMF. (f) BayesSparse. (g) HySure. (h) APNN. (i) DRPNN. (j) HyperPNN1. (k) HyperPNN2. (l) HSpeNet1. (m) HSpeNet2. (n) IC-SA-DyPNN. (o) IC-AT-DyPNN. (p) IC-DC-DyPNN. (q) EC-SA-DyPNN. (r) EC-AT-DyPNN. (s) EC-DC-DyPNN.

and 16-GB RAM. For the CNN-based methods, three different regions of each dataset are selected for training, validation, and testing, respectively. For the training and validation regions, we first perform a downsampling with Gaussian blurring kernels, as mentioned in Section IV-A. The blurred images are then partitioned into partially overlapping samples with the windows size of 20×20 and the sampling stride equal to 8. Furthermore, these image samples are enhanced through rotation and flip to augment the robustness of the designed models. Finally, we use 12 800 of the image samples as the training set, while 2560 of the image samples as the validation set. In addition, the iteration number in the training phase is set as 2.5×10^5, with the batch size of 64. The initial learning rate is empirically set to 5×10^{-5}, and it decays to half every 10^5 iterations. For the optimizations of the proposed DyPNNs, the Adam algorithm is applied to minimize the ℓ_1 loss function. Moreover, the number of scene-sensitive modules in the proposed DyPNNs is empirically set to 3.

D. Experimental Results

Table II reports the quality assessment results on the Washington DC Mall dataset. The best numeric results for each metric are shown in bold, and the second-best ones are underlined. As can be seen, GSA and SFIM achieve

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TABLE III
QUALITY METRICS FOR THE PAVIA UNIVERSITY DATASET

Ours	*IC-SA-DyPNN*	*IC-AT-DyPNN*	*IC-DC-DyPNN*	*EC-SA-DyPNN*	*EC-AT-DyPNN*	*EC-DC-DyPNN*
CC(↑)	0.9669	**0.9673**	0.9671	0.9633	0.9671	0.9657
SAM(↓)	3.4793	**3.4690**	3.5168	3.6315	3.4718	3.5179
ERGAS(↓)	2.1494	2.1414	2.1537	2.2382	**2.1397**	2.1782
RMSE(↓)	0.0221	**0.0220**	0.0221	0.0235	0.0222	0.0226
$Q2^n$(↑)	0.9415	0.9424	**0.9430**	0.9371	0.9421	0.9404
Exisiting	Interpolation	GSA	SFIM	CNMF	BayesSparse	HySure
CC(↑)	0.8663	0.9138	0.9157	0.8559	0.9153	0.9040
SAM(↓)	5.8114	5.5828	5.2997	6.8759	6.2104	5.8408
ERGAS(↓)	5.3008	3.7858	3.6653	4.3740	3.7680	3.8176
RMSE(↓)	0.0409	0.0346	0.0341	0.0454	0.0379	0.0390
$Q2^n$(↑)	0.7342	0.8525	0.8582	0.7808	0.8730	0.8430
Exisiting	APNN	DRPNN	HyperPNN1	HyperPNN2	HSpeNet1	HSpeNet2
CC(↑)	0.9414	0.9503	0.9492	0.9477	0.9544	0.9506
SAM(↓)	4.8641	4.2819	4.2873	4.3968	4.0157	4.1092
ERGAS(↓)	2.9685	2.6939	2.7177	2.7281	2.5781	2.6494
RMSE(↓)	0.0289	0.0268	0.0271	0.0275	0.0257	0.0269
$Q2^n$(↑)	0.8971	0.9174	0.9159	0.9153	0.9233	0.9179

TABLE IV
QUALITY METRICS FOR THE HOUSTON DATASET

Ours	*IC-SA-DyPNN*	*IC-AT-DyPNN*	*IC-DC-DyPNN*	*EC-SA-DyPNN*	*EC-AT-DyPNN*	*EC-DC-DyPNN*
CC(↑)	0.9563	0.9542	**0.9567**	0.9559	0.9544	0.9563
SAM(↓)	3.0273	3.0546	3.0203	3.0285	3.0778	**2.9756**
ERGAS(↓)	**2.3827**	2.4583	2.3934	2.4006	2.4321	2.3900
RMSE(↓)	**0.0126**	0.0130	0.0127	0.0127	0.0129	**0.0126**
$Q2^n$(↑)	0.9288	0.9269	0.9297	0.9280	0.9262	**0.9308**
Exisiting	Interpolation	GSA	SFIM	CNMF	BayesSparse	HySure
CC(↑)	0.8571	0.9187	0.9224	0.8426	0.9231	0.9174
SAM(↓)	6.0076	5.0450	5.1593	7.1996	4.5611	5.2151
ERGAS(↓)	5.5712	3.4906	3.4764	4.6836	3.1292	3.4111
RMSE(↓)	0.0293	0.0186	0.0186	0.0251	0.0167	0.0181
$Q2^n$(↑)	0.7060	0.8599	0.8435	0.7631	0.8698	0.8546
Exisiting	APNN	DRPNN	HyperPNN1	HyperPNN2	HSpeNet1	HSpeNet2
CC(↑)	0.9413	0.9521	0.9503	0.9512	0.9522	0.9519
SAM(↓)	3.7285	3.2793	3.2692	3.2609	3.0973	3.1220
ERGAS(↓)	2.7310	2.4950	2.5260	2.5103	2.5255	2.5101
RMSE(↓)	0.0142	0.0132	0.0131	0.0131	0.0134	0.0133
$Q2^n$(↑)	0.8986	0.9218	0.9197	0.9209	0.9210	0.9183

similar numeric results, while the latter presents a slight advantage. Among three VO-based methods, CNMF gets the most inferior results, where great spatial and spectral distortions are indicated. BayeSparse gains the better performance compared with other traditional pansharpening methods. Close inspection indicates that CNN-based methods yield overall better numeric results than the traditional ones, although APNN presents relative low value on CC metric. Among them, HyperPNNs (i.e., HyperPNN1 and HyperPNN2) perform well on SAM metric, whereas they emerge with relative poor CC metric. In comparison, DRPNN can achieve relatively high value on CC metric, indicating its appreciable spatial fidelity. Besides, HSpeNets (i.e., HSpeNet1 and HSpeNet2) produce the more comprehensively good results in terms of CC and SAM metrics. Notably, our proposed DyPNNs implements significant improvements over all of the comparative methods

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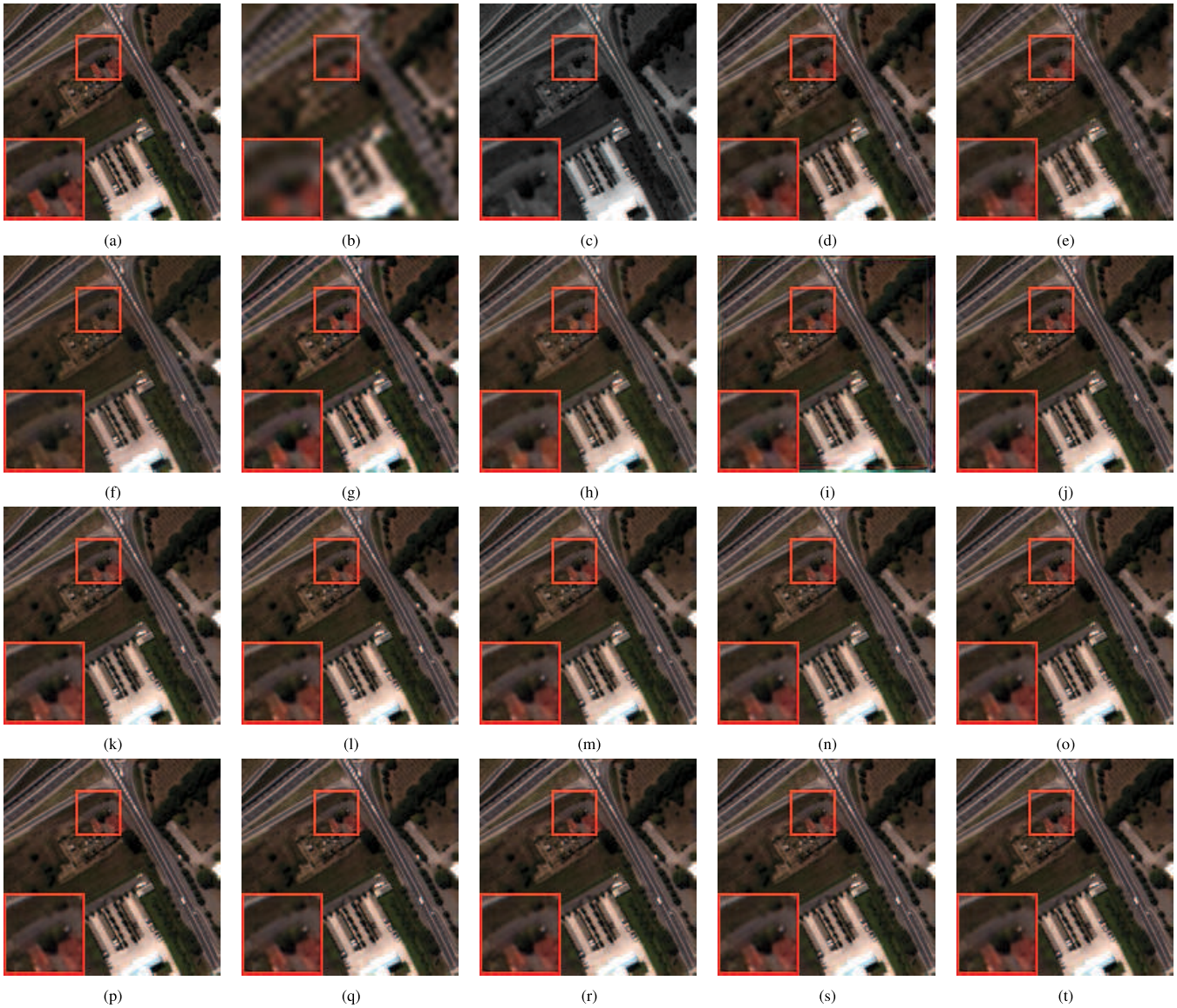

Fig. 8. Pansharpening results of various methods on the Pavia University dataset. (a) Ground truth. (b) Interpolation. (c) PAN. (d) GSA. (e) SFIM. (f) CNMF. (g) BayesSparse. (h) HySure. (i) APNN. (j) DRPNN. (k) HyperPNN1. (l) HyperPNN2. (m) HSpeNet1. (n) HSpeNet2. (o) IC-SA-DyPNN. (p) IC-AT-DyPNN. (q) IC-DC-DyPNN. (r) EC-SA-DyPNN. (s) EC-AT-DyPNN. (t) EC-DC-DyPNN.

on almost all the considered metrics, especially the impressive gains on CC and SAM metrics, which imply the excellent spatial reconstructions and spectral fidelities of our methods. Therein, IC-DC-DyPNN performs best on CC metric as well as the global indicators, including ERGAS and $Q2^n$, while EC-AT-DyPNN is characterized by the highest SAM value. On the whole, all results in Table II demonstrate the superior pansharpening performance of the our DyPNNs.

Fig. 6 visually shows the pansharpening results of different methods on the Washington DC Mall dataset, which provides a visual inspection as a supplementary comparison. As presented, SFIM incurs an oversmoothing pansharpening result, which appears blur and artifact obviously. CNMF, BayesSparse, and HySure can achieve relative appreciable visual inspection results, whereas spectral distortions in these results could not be neglected, such as the poor spectral restore in the zoomed-in areas. Among the CNN-based methods, the pansharpening result of APNN shows pseudo-ribbon at its edge, which may be caused by the use of no-padding convolutional kernels in this method. Pansharpening results of the other CNN-based methods seem indistinguishable; here, the error images of different methods are provided in Fig. 7 for further analyses. It can be seen that the error images of CNN-based methods show smaller amplitudes compared with those of traditional pansharpening methods. In addition, the error images of the proposed DyPNNs present sparser areas with high amplitudes, such as the top-right and bottom-middle areas in these images, which implies that pansharpening results of our methods are more similar to the ground truth, and thus, the superior spatial and the spectral restore performances of our DyPNNs can be verified effectively.

For the Pavia University dataset, quality assessment results of different methods are gathered in Table III. Consistent with the previous analysis for the Washington DC Mall dataset,

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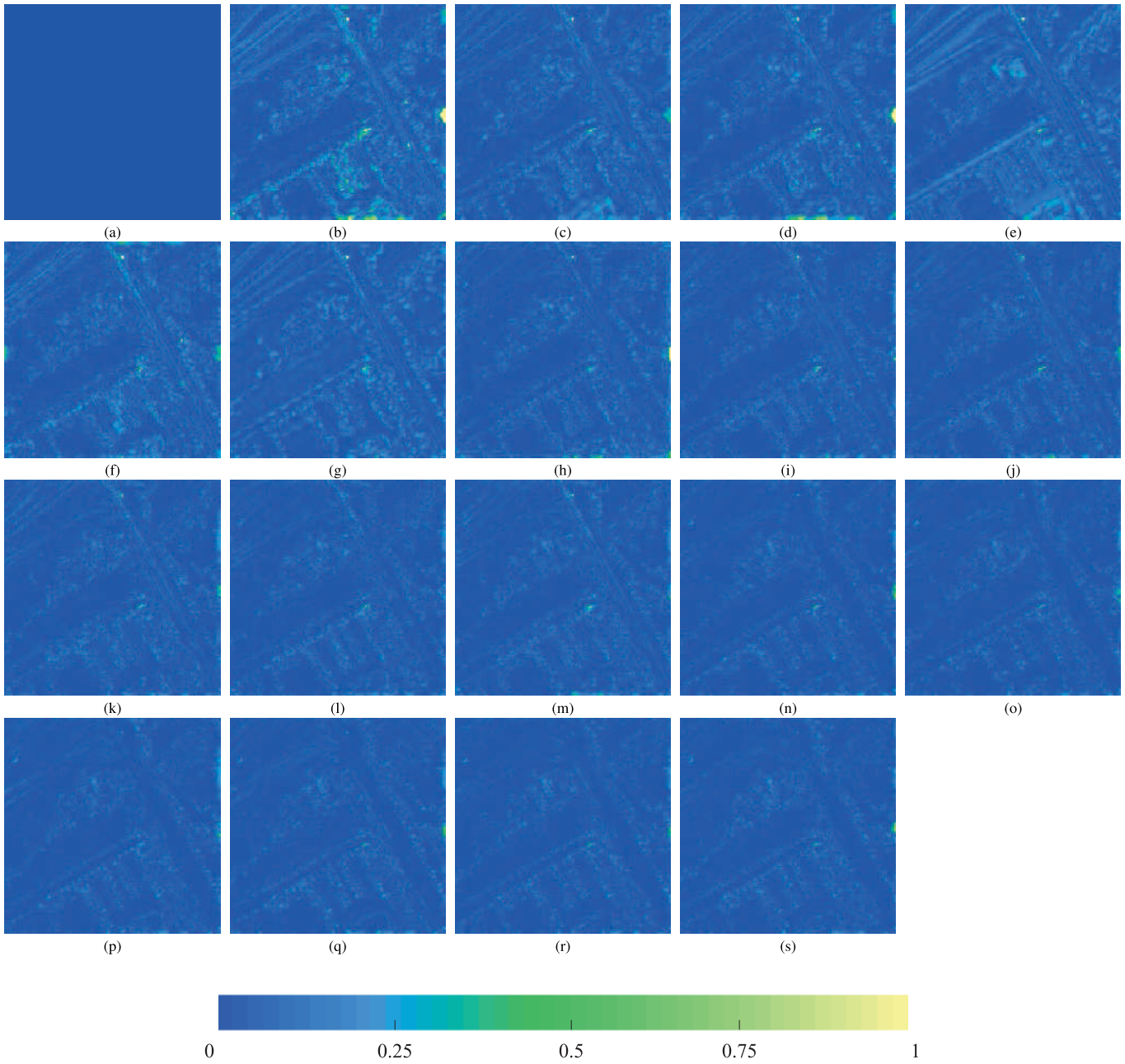

Fig. 9. Error images of various methods on the Pavia University dataset. (a) Ground truth. (b) Interpolation. (c) GSA. (d) SFIM. (e) CNMF. (f) BayesSparse. (g) HySure. (h) APNN. (i) DRPNN. (j) HyperPNN1. (k) HyperPNN2. (l) HSpeNet1. (m) HSpeNet2. (n) IC-SA-DyPNN. (o) IC-AT-DyPNN. (p) IC-DC-DyPNN. (q) EC-SA-DyPNN. (r) EC-AT-DyPNN. (s) EC-DC-DyPNN.

the proposed DyPNNs achieve better numeric results compared with all the other experimental methods. Especially, IC-AT-DyPNN performs best on CC, SAM, and RMSE metrics, which indicates the appreciable advantage of this method, while IC-DC-DyPNN and EC-AT-DyPNN yield the second-best results. The excellent performance of our DyPNNs is also quantitatively verified on this dataset.

Fig. 8 visually exhibits the pansharpening results on the Pavia University dataset. Similar to the result display of the Washington DC Mall dataset, pansharpened results of all compared methods are gathered in Fig. 8. It is observed that the pansharpened image of the CNMF method has serious spectral distortions. Both PCA and SFIM have oversmoothing pansharpening results, with obvious blurs and artifacts. Possibly due to the no-padding convolutional kernels, the pansharpening result of APNN can see the pseudo-ribbon at its edge. Likewise, error images are given for a more clearer visual comparison, as shown in Fig. 9. It can be seen that the proposed DyPNNs presents smaller differences at almost all of spatial positions, which effectively verifies the spatial maintaining abilities of our DyPNNs.

Table IV reports the quality assessment results on the Houston dataset. The similar conclusion as those on the Washington DC Mall dataset and the Pavia University dataset can be drawn from it. We can see that EC-DC-DyPNN performs well on SAM, RMSE, and $Q2^n$ metrics, while IC-SA-DyPNN

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TABLE V
QUALITY METRICS FOR THE MERCED DATASET

Ours	*IC-SA-DyPNN*	*IC-AT-DyPNN*	*IC-DC-DyPNN*	*EC-SA-DyPNN*	*EC-AT-DyPNN*	*EC-DC-DyPNN*
CC(↑)	0.9739	0.9742	0.9726	0.9726	**0.9755**	0.9677
SAM(↓)	2.6719	2.7701	2.6839	2.7033	**2.6500**	2.8341
ERGAS(↓)	3.8628	3.8476	3.8869	3.9486	**3.7521**	4.1349
RMSE(↓)	0.0167	**0.0166**	0.0171	0.0171	**0.0161**	0.0187
$Q2^n$(↑)	0.9485	0.9498	0.9478	0.9460	**0.9519**	0.9420
D_λ(↓)	0.0237	0.0268	0.0266	0.0181	0.0208	0.0246
D_s(↓)	0.0100	0.0122	0.0097	**0.0065**	0.0094	0.0146
QNR(↑)	0.9665	0.9614	0.9640	**0.9755**	0.9701	0.9611
Existing	Interpolation	GSA	SFIM	CNMF	BayesSparse	HySure
CC(↑)	0.9570	0.9570	0.9349	0.7807	0.9521	0.6392
SAM(↓)	3.4553	3.4558	3.7041	4.4793	4.6978	13.2422
ERGAS(↓)	5.1410	5.1404	6.5352	10.0028	6.0189	18.3934
RMSE(↓)	0.0215	0.0215	0.0265	0.0489	0.0270	0.0874
$Q2^n$(↑)	0.9032	0.9032	0.8514	0.6590	0.8998	0.4953
D_λ(↓)	**0.0069**	0.1522	0.0499	0.0951	0.0416	0.0791
D_s(↓)	0.0278	0.0298	0.0972	0.2300	0.0265	0.0697
QNR(↑)	0.9655	0.8225	0.8577	0.6968	0.9329	0.8567
Existing	APNN	DRPNN	HyperPNN1	HyperPNN2	HSpeNet1	HSpeNet2
CC(↑)	0.9707	0.9496	0.9701	0.9696	0.9471	0.9588
SAM(↓)	2.9831	4.2625	3.0734	3.0792	2.8558	3.2254
ERGAS(↓)	4.2426	5.7639	4.1892	4.2183	5.1949	5.0880
RMSE(↓)	0.0170	0.0232	0.0178	0.0179	0.0254	0.0214
$Q2^n$(↑)	0.9384	0.9018	0.9401	0.9392	0.8926	0.9163
D_λ(↓)	0.0274	0.0437	0.0354	0.0295	0.0193	0.0285
D_s(↓)	0.0099	0.0355	0.0109	0.0131	0.0403	0.0197
QNR(↑)	0.9630	0.9224	0.9541	0.9578	0.9412	0.9524

is competitive on CC metric as well as the global ERGAS and RMSE metrics, and IC-DC-DyPNN is characterized by the highest value of CC metric. Overall, several DyPNNs we propose are able to achieve overall high-quality pansharpening results.

E. Full-Resolution Experiment on Real Hyperspectral Dataset

In order to better validate the generalization ability of the proposed DyPNNs in preserving spatial and spectral resolutions, full-resolution experiments are conducted on the real HS dataset, i.e., the Merced dataset. For a comprehensive comparison, the experimental results are shown in both quantitative and visual fashions.

Table V exhibits the quantitative results, including the no-reference metrics D_λ, D_s, and quality with no-reference (QNR), which are usually specialized for the full-resolution evaluation [49], and additionally gives the results from metrics in the reduced-resolution level. As observed, DyPNNs we designed tend to achieve a better performance compared with other methods, which verifies the superiority of the learn-to-learn strategy. It can be seen that the EC-SA-DyPNN achieves the best scores on D_λ, D_s, and QNR among all the methods (except the Interpolation), since the learn-to-learn strategy helps the CNNs adapt the untrained data. It is noteworthy that the APNN also shows good scores on the no-reference metrics with the help of the target adaptive technique, in which the CNN is fine-tuned with the oncoming data to solve the discrepancy between the training data and the panshrapening data and, therefore, could be somewhat adaptive to the oncoming data needed to handle.

In addition, the visual comparison is provided for intuitive understanding, as shown in Fig. 10. As observed, our DyPNNs preserve abundant contour information and produce no artifacts. Furthermore, the visual results achieved by our DyPNNs look clearer and more natural. Intuitively, our methods better recover the details and better retain the fidelity of the spectral information compared with the other methods.

F. Ablation Study

For better understanding how our learn-to-learn-based DyPNNs work, we perform an ablation study on the Pavia University dataset. Since the MSDB subnetwork and the modulating operation $\mathcal{G}(\cdot)$ are cores to realize the learn-to-learn strategy, these blocks are removed from DyPNNs to form an ablation pansharpening network, which is used in the ablation comparison. In Fig. 11, the ablation network and six DyPNNs are compared. As observed, although with exactly the same architecture as DyPNNs except for MSDB and $\mathcal{G}(\cdot)$,

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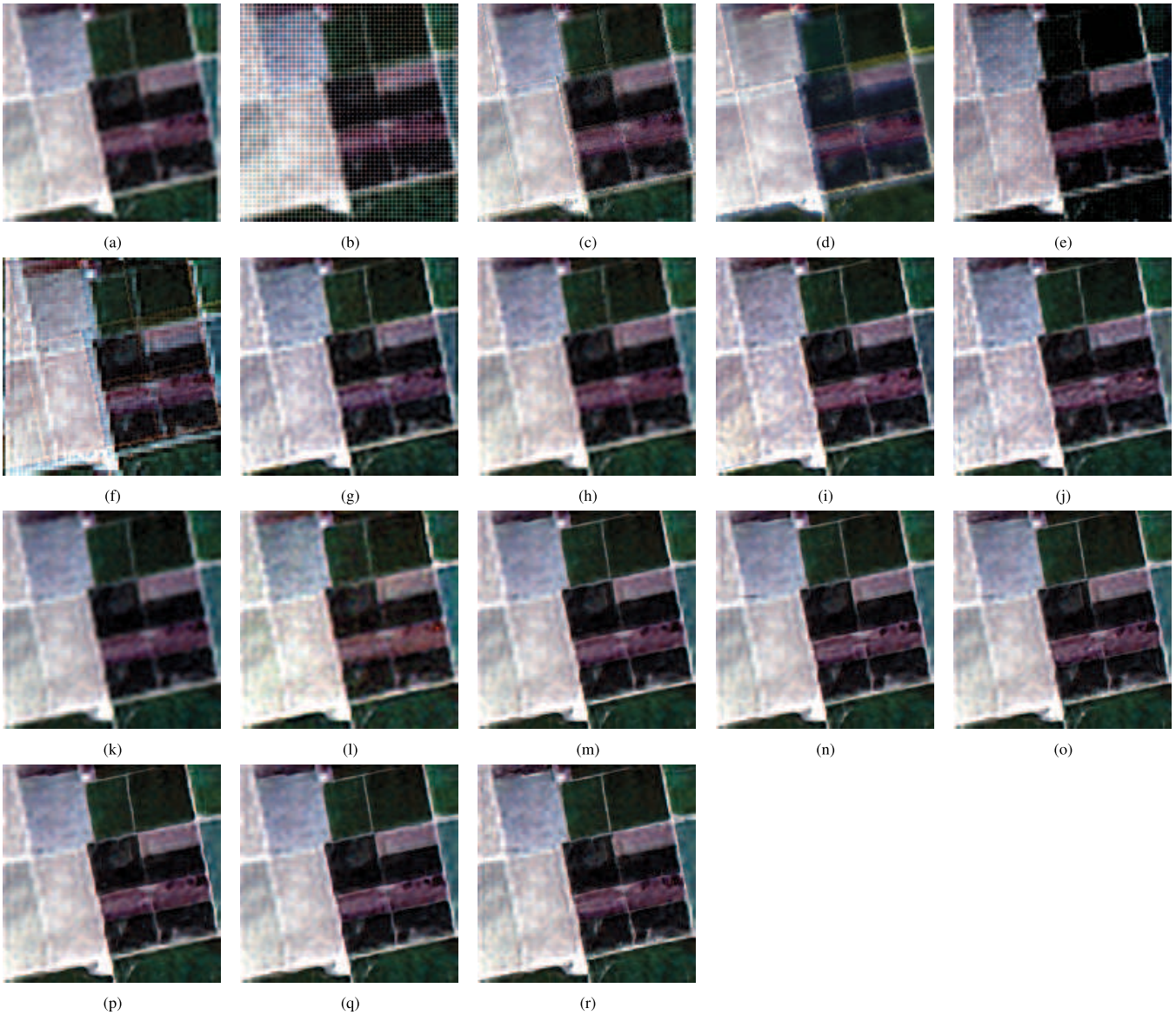

Fig. 10. Full-resolution Pansharpening results of various methods on the Merced dataset. (a) Interpolation. (b) GSA. (c) SFIM. (d) CNMF. (e) BayesSparse. (f) HySure. (g) APNN. (h) DRPNN. (i) HyperPNN1. (j) HyperPNN2. (k) HSpeNet1. (l) HSpeNet2. (m) IC-SA-DyPNN. (n) IC-AT-DyPNN. (o) IC-DC-DyPNN. (p) EC-SA-DyPNN. (q) EC-AT-DyPNN. (r) EC-DC-DyPNN.

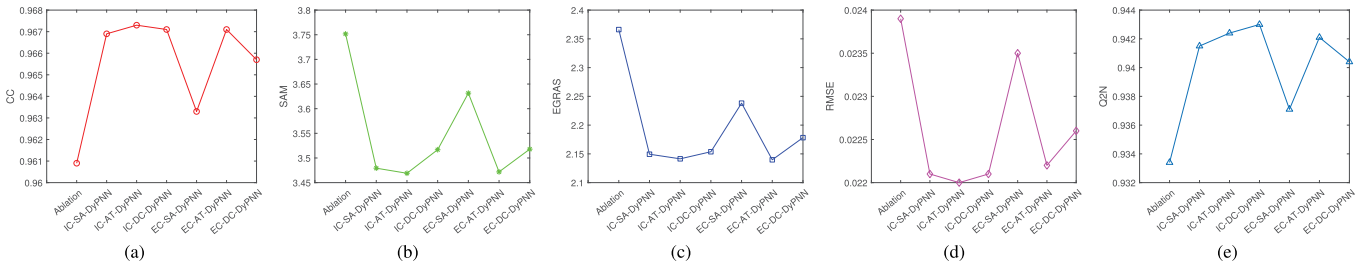

Fig. 11. Ablation study on the Pavia University dataset. (a) CC. (b) SAM. (c) EGRAS. (d) RMSE. (e) $Q2^n$.

the ablation network yields worse performances. All the five metrics of the ablation network are obviously inferior to the other six DyPNNs. This phenomenon truly indicates the effectiveness of the MSDB component and modulating operation unit $\mathcal{G}(\cdot)$ in our DyPNNs and, thus, the effectiveness of the learn-to-learn strategy in designing dynamic pansharpening.

Summing up all the above experimental results, it is suggested that the family of our DyPNNs is able to achieve overall excellent pansharpening performances in terms of both numeric evaluation and visual inspection, and thus, the strategy using learn-to-learn to guide the design of dynamic pansharpening is effective.

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V. Conclusion

Considering spatial variation occurred in the task of HS pansharpening, in this article, we seek to develop DyPNN methodology. First, we constructed a learn-to-learn-oriented pansharpening paradigm to accommodate the spatially adaptive adjustments during HS panshrapening tasks. Then, following the paradigm, we proposed two groups of DyPNNs, which comprises six specific DyPNNs. We tested our DyPNNs on several real datasets. The quantitative and visual evaluation results on four real-world datasets suggested the superior pansharpening performance of our proposed DyPNNs in terms of spatial and spectral restorations and, thus, the effectiveness of the learn-to-learn strategy in guiding the design of dynamic pansharpening. In the future, learn-to-learn could be considered to guide designing more effective pansharpening methods.

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